

Q-PUFAuth: Quantum-Resistant PUF-Based Authentication and Key Agreement for Secure Smart Grids

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Abstract. The increasing reliance on smart grid technologies necessitates robust authentication and key agreement mechanisms to secure critical infrastructure against cyber threats. Traditional authentication schemes such as LAP-SG provide lightweight authentication but suffer from vulnerabilities, including susceptibility to quantum attacks and weak physically unclonable function (PUF) implementations. In response, this paper presents an enhanced version of the LAP-SG protocol with strengthened security features and improved efficiency. Our proposed protocol integrates quantum-resistant cryptographic mechanisms, leveraging lattice-based encryption to withstand attacks from quantum adversaries. Additionally, we enhance PUF challenge-response mechanisms to mitigate cloning and side-channel attacks. A formal security verification of our protocol is conducted using ProVerif and the Real-OR-Random (ROR) model. The proposed approach ensures mutual authentication, forward secrecy, resistance to impersonation and replay attacks, and scalability for large-scale smart grid deployments. This work contributes a quantum-secure and lightweight authentication framework for future smart grid security needs.

Keywords: Smart Grids · Quantum-Resistant Security · Physically Unclonable Functions (PUF) · Mutual authentication · Key agreement.

1 Introduction

With the rise of smart grid technologies, robust authentication and key agreement protocols are essential to protect infrastructure against cyber threats. While smart meters and intelligent control centers enhance efficiency, they also introduce vulnerabilities such as unauthorized access, replay attacks, and quantum threats.

LAP-SG [11] was proposed as a lightweight authentication protocol, minimizing computational costs for resource-limited devices. However, it presents several limitations, including weak PUF usage, lack of formal security analysis, and susceptibility to quantum attacks. Given the evolution of quantum computing, traditional schemes like RSA and ECC are no longer sufficient.

To address these challenges, we propose Q-PUFAuth: an enhanced version of LAP-SG integrating lattice-based cryptography (Kyber, Dilithium) for quantum resistance, and strong PUF-based authentication. Our approach includes formal verification using ProVerif and the Real-or-Random model, ensuring mutual authentication, forward secrecy, and protection against common attacks.

The paper is organized as follows: Section 2 reviews related works. Section 3 introduces preliminaries. Section 4 describes Q-PUFAuth. Section 5 presents performance comparisons. Section 6 provides a security analysis, and Section 7 concludes the paper.

2 Related Work

Recent years have witnessed the development of various authentication and key agreement protocols tailored for smart grid environments. Gope and Sikdar [7] introduced a PUF-based key agreement method that aimed to provide enhanced privacy-aware authentication. While their protocol leverages Physically Unclonable Functions (PUFs) to establish unique device identities, subsequent analysis by Braeken et al. [4] uncovered vulnerabilities, particularly in the form of replay attacks and weak challenge-response mechanisms. Additionally, the cryptographic primitives employed in their scheme remain susceptible to quantum adversaries, rendering them inadequate for long-term security in post-quantum scenarios.

Another approach [15] focused on lightweight and anonymous authentication for smart power grids. This protocol sought to minimize computational overhead while maintaining security. However, it remains vulnerable to impersonation attacks, and its forward secrecy is not robust enough to prevent retrospective decryption of past communications should a session key be compromised. Moreover, the reliance on low-power smart meters introduces a risk of denial-of-service (DoS) attacks due to constrained computational capabilities.

In response to the emerging quantum threat landscape, researchers have begun to explore quantum-resistant authentication mechanisms. A notable scheme integrated lattice-based cryptography [5] to enhance security against both classical and quantum attacks. Despite its advantages, this approach introduced significant computational overhead, which can be prohibitive for the resource-constrained devices deployed in smart grids. Additionally, improper parameter selection may expose the system to side-channel attacks, weakening its resilience.

The certificate-less authentication and key agreement protocol (ECL-AKA) [9] has also been proposed to mitigate security risks in smart grid communications. By eliminating the need for traditional public key infrastructure, this protocol reduces computational costs and enhances efficiency. However, it introduces the key escrow problem, where a central authority becomes a single point of failure. Furthermore, potential man-in-the-middle attacks may still compromise its security due to reliance on intermediate nodes for key verification. Despite improvements over classical schemes, its resistance to quantum attacks remains unverified in formal cryptographic models. Identity-based cryptographic protocols combined with elliptic curve cryptography (ECC) [2, 10, 16] have also been explored as an alternative lightweight authentication solution. While these schemes offer reduced computational complexity, they may still be susceptible to brute-force attacks, particularly as computational power continues to increase. Moreover, in cases where private keys are compromised, an attacker could forge authentication messages, leading to identity theft and impersonation attacks. Additionally, dynamic group key management presents a challenge in large-scale deployments, potentially leading to synchronization issues and delays in authentication processes.

Although these protocols have introduced notable advancements, they continue to exhibit security vulnerabilities that must be addressed. The proposed protocol in this paper aims to overcome these limitations by implementing stronger PUF mechanisms to prevent cloning attacks, integrating quantum-resistant cryptographic techniques to ensure long-term security, improving nonce and session key management to enhance forward secrecy, and utilizing formal verification tools such as ProVerif to rigorously analyze security properties. Through these enhancements, our protocol provides a more robust, scalable, and efficient authentication framework for smart grid applications.

3 Preliminaries

This section provides an overview of the fundamental cryptographic techniques used in our proposed authentication and key agreement protocol by integrating Physically Unclonable Functions (PUFs), Kyber, and Dilithium. PUFs [12], which are hardware-based security primitives exploiting manufacturing variations, generate unique Challenge-Response Pairs (CRPs) for device authentication and key generation, and are embedded in each smart meter for secure identification. To establish a session key resistant to quantum attacks, our protocol employs Kyber [3], a post-quantum key encapsulation mechanism based on the Learning With Errors (LWE) problem [13] over Module Lattices [1], operating through key generation, encapsulation, and decapsulation algorithms. Complementing this, message authentication between smart meters and the control center is ensured using Dilithium [6], a post-quantum digital signature scheme built upon Module LWE and Module SIS problems, which performs secure key generation, signature creation, and verification to maintain message integrity.

4 Proposed Approach

This section presents the design of our proposed authentication and key agreement scheme, Q-PUFAuth. It includes both the system model and the threat model that guide the security assumptions. These models establish the context in which Q-PUFAuth operates and define the adversary capabilities considered during protocol design.

4.1 System Model

Our improved authentication and key agreement protocol is designed for smart grid environments, which consist of the following key entities:

- **Trusted Authority (TA):** The TA is responsible for initializing the system by generating cryptographic parameters and registering smart grid devices. It provides each device with a unique identity and ensures secure enrollment.
- **Smart Meters (SMs):** These are resource-constrained edge devices deployed in smart grid infrastructures to monitor energy consumption and communicate with control centers.
- **Control Center (CC):** The CC manages smart meters, processes authentication requests, and establishes secure communication channels with authenticated devices.

The proposed system incorporates Physically Unclonable Functions (PUFs) to provide unique device identities and prevent cloning attacks. It also integrates lattice-based cryptographic techniques for quantum-resistant authentication and key agreement. Secure communication is ensured by generating a session key shared between the smart meter and the control center.

4.2 Threat Model

The security of smart grid communication is threatened by a wide range of adversaries. We consider the following threat model in our analysis:

- **Adversary Capabilities:** The adversary $\mathcal{A}$ has full control over the communication channel, allowing it to intercept, modify, replay, or inject messages. $\mathcal{A}$ may attempt to impersonate legitimate entities, extract cryptographic keys, or launch quantum-based attacks.

- **Impersonation Attacks:** The adversary attempts to impersonate a legitimate smart meter or control center by forging authentication messages.
- **Man-in-the-Middle (MitM) Attacks:** $\mathcal{A}$ intercepts and modifies communication between smart meters and control centers to manipulate authentication and key exchange.
- **Replay Attacks:** Previously exchanged authentication messages may be replayed to gain unauthorized access.
- **Quantum Adversary Model:** We assume an attacker with access to quantum computing resources that can break traditional cryptographic schemes (e.g., RSA, ECC). Our protocol counters this threat by employing lattice-based cryptographic primitives.
- **PUF-Based Attacks:** The adversary may attempt to exploit weaknesses in the PUF implementation, such as machine learning-based modeling attacks, to predict challenge-response pairs (CRPs).
- **Key Compromise Impersonation (KCI) Attacks:** If a smart meter’s secret key is compromised, the adversary may attempt to impersonate other entities.

To mitigate these threats, Q-PUFAuth ensures mutual authentication, forward secrecy, and resistance to key-compromise impersonation by integrating quantum-resistant cryptography, secure PUF mechanisms, and formal verification using ProVerif.

4.3 Protocol Description

This subsection presents Q-PUFAuth, our enhanced authentication and key agreement protocol designed to address the vulnerabilities of LAP-SG. It integrates Kyber and Dilithium to achieve quantum resistance, and reinforces PUF security through robust challenge-response mechanisms. CRPs are hashed and encrypted to resist machine learning attacks. The protocol ensures mutual authentication, forward secrecy, and protection against replay attacks using nonces and timestamps. Its efficient key management also supports scalability for large-scale smart grid environments. Table 1 summarizes the notations used.

Table 1: List of Symbols and Their Descriptions

Symbol	Description
ID_{SM}, ID_{CC}	Identity of the Smart Meter (SM) and Control center (CC)
PUF	Physically Unclonable Function
H_{PUF}	Helper data used to correct PUF responses
R_{PUF}	Response generated by the PUF
$H(.)$	One-way hash function
Ch_{CC}	Challenge issued by the Control Center
$Resp_{SM}$	Response from the Smart Meter to the challenge
σ_{SM}, σ_{CC}	Signature generated by the SM, the CC respectively using Dilithium
$SK_{session}$	Session key derived using Kyber KEM
$PK_{SM}^{Dilithium}$	Public key of the Smart Meter for Dilithium signatures
$PK_{CC}^{Dilithium}$	Public key of the Control Center for Dilithium signatures
SK_{SM}^{Kyber}	Secret key of the Smart Meter for Kyber key exchange
PK_{SM}^{Kyber}	Public key of the Smart Meter for Kyber key exchange
$Ciphertext_{Kyber}$	Encapsulated session key using Kyber

System Initialization Phase A Trusted Authority (TA) manages key generation, authentication, and PUF enrollment. The initialization follows these steps:

- Step 1: TA Key Generation. The TA generates a master key pair: (PK_{TA}, SK_{TA}) using Dilithium for signing certificates. The TA's public key (PK_{TA}) is pre-distributed to all entities.
- Step 2: Control Center Registration. The control center generates: $(PK_{CC}^{Kyber}, SK_{CC}^{Kyber})$ for Kyber key exchange and $(PK_{CC}^{Dilithium}, SK_{CC}^{Dilithium})$ for Dilithium authentication. It sends a certificate request to TA:

$$M_{CC} = (ID_{CC}, PK_{CC}^{Kyber}, PK_{CC}^{Dilithium}, \sigma_{CC}) \quad (1)$$

where $\sigma_{CC} = \text{Sign}_{SK_{CC}^{Dilithium}}(ID_{CC}, PK_{CC}^{Kyber}, PK_{CC}^{Dilithium})$.

The TA verifies the request and issues a certificate:

$$Cert_{CC} = \text{Sign}_{SK_{TA}}(ID_{CC}, PK_{CC}^{Kyber}, PK_{CC}^{Dilithium}) \quad (2)$$

The control center stores its certificate.

- Step 3: Smart Meter Registration & PUF Enrollment. Each smart meter (SM) generates: $(PK_{SM}^{Dilithium}, SK_{SM}^{Dilithium})$ for authentication. During the PUF enrollment process, the TA issues multiple random challenges Ch_i to the PUF inside the smart meter. In response, the PUF generates outputs $R_{PUF,i}$, which may vary slightly due to noise. To correct errors in future responses, a helper data algorithm, such as a fuzzy extractor or BCH error correction, generates helper data $H_{PUF,i}$. The TA securely stores $H_{PUF,i}$ but never retains $R_{PUF,i}$ to maintain security. The SM sends a registration request:

$$M_{SM} = (ID_{SM}, PK_{SM}^{Dilithium}, H(R_{PUF}), \sigma_{SM}) \quad (3)$$

where $\sigma_{SM} = \text{Sign}_{SK_{SM}^{Dilithium}}(ID_{SM}, H(R_{PUF}))$. The TA verifies the request and issues a certificate:

$$Cert_{SM} = \text{Sign}_{SK_{TA}}(ID_{SM}, PK_{SM}^{Dilithium}) \quad (4)$$

The SM stores its certificate.

- Step 4: Secure Deployment. The Control Center (CC) and Smart Meter (SM) establish secure communication using their TA-certified public keys. During authentication, the PUF response is first corrected with the help of associated helper data before being hashed and sent for verification. The TA is only involved when revocation or re-certification is required.

Login and Key Agreement Phase

Step 1: Smart Meter Login

The smart meter initiates authentication by sending an encrypted PUF response along with a timestamp to prevent replay attacks:

$$M_1 = (ID_{SM}, Enc_{PK_{CC}^{Kyber}}(R_{PUF}||T_1), \sigma_{SM}) \quad (5)$$

where ID_{SM} represents the smart meter identity, $Enc_{PK_{CC}^{Kyber}}(R_{PUF}||T_1)$ is the PUF response and timestamp encrypted with the control center's Kyber public key, and σ_{SM} is the signature generated using Dilithium.

Step 2: Control Center Verification and Key Exchange

Upon receiving the message, the control center decrypts $Enc_{PK_{CC}^{Kyber}}(R_{PUF}||T_1)$ to retrieve R_{PUF} , validates the timestamp T_1 to prevent replay attacks, and verifies the signature using $PK_{SM}^{Dilithium}$. If the authentication is successful, the control center generates a randomized challenge:

$$Ch_{CC} = H(N_{CC}||ID_{SM}) \quad (6)$$

where N_{CC} is a random nonce. It then encapsulates a session key using Kyber KEM and sends the following message:

$$M_2 = (Ciphertext_{Kyber}, Ch_{CC}, T_2, \sigma_{CC}) \quad (7)$$

where $Ciphertext_{Kyber}$ is the encapsulated session key, Ch_{CC} is the hashed challenge to prevent direct observation, T_2 is the control center timestamp, and σ_{CC} is the signature generated using Dilithium.

Step 3: Smart Meter Response and Mutual Authentication

Upon receiving M_2 , the smart meter verifies σ_{CC} , checks T_2 , and derives the session key from $Ciphertext_{Kyber}$. It then computes a hashed response to Ch_{CC} using the PUF:

$$Resp_{SM} = H(Ch_{CC}||R_{PUF}||SK) \quad (8)$$

The smart meter then sends:

$$M_3 = (H(Resp_{SM}), T_3, \sigma_{SM}) \quad (9)$$

where $H(Resp_{SM})$ ensures protection against machine learning attacks, T_3 is the smart meter timestamp, and σ_{SM} proves response authenticity.

Step 4: Control Center Final Verification and Secure Communication

The control center verifies σ_{SM} and computes:

$$H(Resp_{SM}^*) = H(Ch_{CC}||R_{PUF}||SK) \quad (10)$$

using the stored R_{PUF} and session key. If the computed hash matches $H(Resp_{SM})$, authentication is confirmed. Secure communication is then established through encrypted data transmission:

$$M_{enc} = Enc_{SK}(Data) \quad (11)$$

This phase, as illustrated in Figure 1 ensures secure key agreement and authentication with enhanced security guarantees.

5 Comparison with LAP-SG

5.1 Comparison of Security Features, Computation, and Communication Costs

To evaluate the efficiency of Q-PUFAuth, we compare its security features, as well as its computation and communication costs, with those of LAP-SG. Our protocol introduces a slight increase in communication overhead due to additional security verification but significantly enhances security, particularly against quantum attacks. Although Q-PUFAuth integrates lattice-based operations, its design remains lightweight due to the use of PUFs, which dynamically generate secrets and eliminate the need for storing long-term cryptographic keys, as well as the efficiency of Kyber and Dilithium

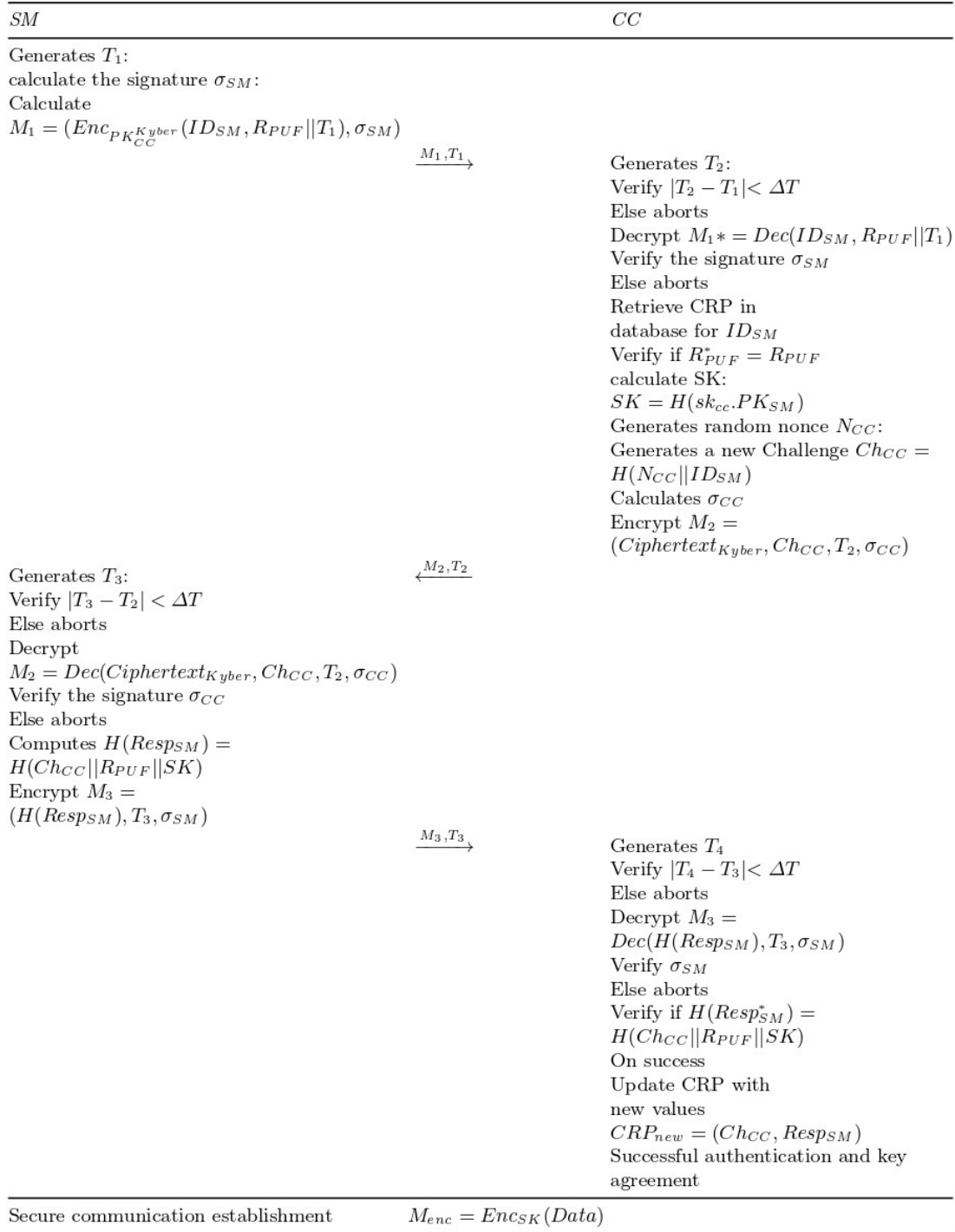


Fig. 1: Login and key agreement phase

operations tailored for constrained devices. The protocol also maintains a compact authentication flow with only three exchanged messages, which balances security and communication overhead. Table 2 presents a comparison between our scheme and LAP-SG.

Table 2: Complexity Comparison between Q-PUFAuth and LAP-SG.

Metric	LAP-SG [11]	Q-PUFAuth
Security against physical attacks	No	Yes (PUF-based)
Number of cryptographic operations	$O(n)$	$O(n)$ (with quantum-resistant ops)
Number of messages exchanged	2	3 (additional security verification)
Key exchange complexity	$O(1)$	$O(1)$ (lattice-based)
PUF evaluations	0	1
Communication overhead (bits)	Low	Slightly higher (due to added security)
Security level	Classical	Quantum-resistant

5.2 Q-PUFAuth: An Improved Yet Distinct Scheme

Although Q-PUFAuth builds upon the foundations of LAP-SG, it introduces significant modifications that go beyond a simple enhancement, making it a distinct authentication and key agreement scheme. While LAP-SG relies on elliptic curve cryptography (ECC) and hash-based authentication, Q-PUFAuth integrates post-quantum security mechanisms, including lattice-based key exchange (Kyber KEM), digital signatures (Dilithium), and Physically Unclonable Functions (PUFs). These modifications not only enhance security against quantum adversaries but also restructure the authentication flow.

Unlike a traditional improvement that refines an existing scheme while preserving its core cryptographic framework, Q-PUFAuth replaces LAP-SG’s underlying security mechanisms, introducing an entirely new authentication approach. The integration of PUFs for device uniqueness and the transition to lattice-based cryptography establish a fundamentally different trust model and security architecture. Consequently, while Q-PUFAuth originates from LAP-SG, its structural changes and quantum-resistant properties qualify it as a new authentication and key agreement scheme rather than a mere enhancement. Table 3 presents a comparison between our scheme and LAP-SG [11].

Table 3: Comparison Between LAP-SG and Q-PUFAuth

Feature	LAP-SG [11]	Q-PUFAuth
Authentication Mechanism	Hash-Based	PUF + Lattice-Based Signatures
Key Exchange	ECC-Based Diffie-Hellman	Kyber KEM (Post-Quantum)
Signature Scheme	ECDSA	Dilithium (Lattice-Based)
Quantum Attacks	Vulnerable	Quantum-Resistant
PUF Utilization	No	Yes (for Device Uniqueness)
Communication Overhead	Moderate	Slightly Higher (Lattice Cryptography)
Computational Cost	Lower (ECC)	Higher (Lattice Operations)
Key Size	Small (ECC Keys)	Larger (Lattice-Based Keys)
Scalability	Suitable for Classical Networks	Future-Proof for Post-Quantum Networks

6 Security Analysis

We evaluate the security of Q-PUFAuth using both formal and informal analysis. Formal methods include verification through ProVerif and a proof using the Real-or-Random (RoR) model. Informally, we analyze the protocol's resistance to various attack vectors common in smart grid environments.

6.1 Formal Security Analysis

ProVerif Security Verification: To formally verify the security of Q-PUFAuth, its authentication and key agreement process was modeled using the ProVerif tool. The verification focuses on three critical security properties: session key, identity protection, and the confidentiality of the Physically Unclonable Function (PUF) response. The ProVerif analysis successfully proves that Q-PUFAuth resists man-in-the-middle (MitM) attacks, replay attacks, and impersonation attacks. The session key remains secret under an adversarial model, confirming the robustness of the lattice-based key exchange and authentication mechanisms. The authentication phase ensures mutual verification between the smart meter and the control center, with PUF-based responses preventing device cloning. and PUF response all return `false` for an attacker, meaning these elements remain confidential.

Real-Or-Random (RoR) Model Proof The Real-or-Random (RoR) model is a well-established security framework for proving the indistinguishability of session keys from truly random values. We adopt the RoR model because it effectively captures adversarial capabilities in realistic attack scenarios, evaluating whether an attacker can differentiate a generated session key from a random one. By demonstrating security in the RoR model, we establish strong guarantees on our protocol's key secrecy, forward secrecy, and resistance against key-compromise impersonation (KCI) attacks. To formally analyze the security of our protocol, we construct a sequence of games:

- **Game 0:** The real protocol execution where the adversary interacts with honest parties. The adversary's advantage is $Adv_{\mathcal{A}}^{(0)}$.
- **Game 1:** Replace session key derivation with a random value. If the adversary distinguishes this, it breaks the LWE problem. The advantage of the adversary is $Adv_{\mathcal{A}}^{(1)} \leq Adv_{LWE}$.
- **Game 2:** Encryptions are replaced with random values under the IND-CPA security assumption. The advantage of the adversary is $Adv_{\mathcal{A}}^{(2)} \leq Adv_{IND-CPA}$.

The total advantage of the adversary is described by equation 12, ensuring session key indistinguishability.

$$Adv_{\mathcal{A}} \leq Adv_{\mathcal{A}}^{(0)} + Adv_{\mathcal{A}}^{(1)} + Adv_{\mathcal{A}}^{(2)} \leq Adv_{LWE} + Adv_{IND-CPA} \quad (12)$$

6.2 Informal Security Analysis

The proposed protocol enhances security against multiple attack vectors commonly observed in smart grid environments:

Resistance to Replay Attacks: The use of timestamps T_1, T_2, T_3 and the verification of the freshness of received messages ensure that authentication messages cannot be maliciously reused. In particular, if an adversary attempts to replay a previously intercepted message (M, T_i, σ) , the

verifier will check the condition $T_{recv} - T_i \leq \Delta T$ and verify the signatures σ_{SM} and σ_{CC} tampered messages are rejected.

Resistance to Man-in-the-Middle Attacks: The mutual authentication process involves challenge-response pairs derived from the PUF. Given a challenge Ch_{CC} , the response $Resp_{SM} = PUF(Ch_{CC})$ is unique and unpredictable, preventing an adversary from injecting falsified responses. Moreover, encrypted session keys $Ciphertext_{Kyber}^{old}$ ensure message integrity and confidentiality. The inclusion of a digital signatures σ_{CC} and σ_{SM} further guarantees authenticity.

Resistance to Impersonation Attacks: An adversary attempting to impersonate a legitimate entity must generate a valid response $Resp_{SM}$ for a given challenge Ch_{CC} . Since the PUF output is not stored and is unique per device, an attacker without physical access to the device cannot compute $Resp_{SM}$, making impersonation infeasible. Additionally, the signatures σ_{CC} and σ_{SM} ensures that only legitimate entities can generate valid messages.

Resistance to Key Compromise Attacks: Even if an adversary gains access to previous session keys $Ciphertext_{Kyber}^{old}$, they cannot derive future session keys due to the inclusion of fresh challenges and responses in each session. The dynamic generation of challenges $Ch_{CC} = H(N_{CC} || ID_{SM})$ ensures forward secrecy, preventing key reuse. The digital signature mechanism further strengthens security by ensuring that only authenticated entities can establish session keys.

Resistance to Physical Attacks: Q-PUFAuth strengthens physical attack resistance by leveraging the intrinsic properties of PUFs. Unlike stored secret keys, PUF responses are generated dynamically based on the unique physical characteristics of each device, which makes them highly resistant to physical extraction via invasive or semi-invasive methods.

Quantum Resistance: The security of Q-PUFAuth is enhanced by its reliance on lattice-based cryptography, specifically the Kyber key exchange mechanism and Dilithium digital signatures, which are both designed to withstand quantum attacks. Unlike traditional public-key cryptosystems that rely on factorization or discrete logarithm problems, lattice-based schemes remain secure against quantum adversaries. The use of Kyber ensures secure key encapsulation, while Dilithium provides strong authentication and message integrity resistant to Shor's [14] and Grover's [8] algorithms.

Resistance to Side-Channel Attacks: Q-PUFAuth provides resilience against side-channel attacks by leveraging the inherent properties of PUFs. Since PUF responses are not stored and are dynamically generated based on physical variations, they do not create exploitable static keys. Additionally, power and timing analysis attacks are mitigated due to the unpredictable nature of PUF evaluations. Countermeasures such as controlled PUFs (C-PUFs) and masking techniques further reduce the risk of leakage through side-channel observations.

7 Conclusion

In this work, we presented an improved authentication and key agreement protocol for smart grids, addressing the security weaknesses of LAP-SG. Our approach integrates quantum-resistant cryptographic mechanisms, enhanced PUF-based authentication, and formal security verification.

Q-PUFAuth offers a well-balanced design that combines robust quantum-resistant security features with acceptable overhead in terms of computation and communication. Although the use of lattice-based primitives introduces higher complexity compared to classical schemes like LAP-SG, the protocol remains suitable for deployment in resource-constrained smart grid environments. Through rigorous security analysis, we demonstrated that our protocol offers superior protection

against various cyber threats, including quantum adversaries, impersonation attacks, and replay attacks.

Future research directions include further optimizations to enhance efficiency, exploration of additional post-quantum cryptographic techniques, and real-world implementation to validate practical feasibility.

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Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

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